The Conic Sections Invariant Theory

Wayne State University, MAT 7470, Winter 2024. Amer Rawashdeh, hf1056@wayne.edu

The general conic section is the zero set of the polynomial

$$p(x,y) = ax^2 + 2bxy + cy^2 + 2dx + 2ey + f. \tag{$*$}$$

Since it is the zero set, that would mean

$$\lambda p(x,y) = a\lambda x^2 + 2b\lambda xy + c\lambda y^2 + 2d\lambda x + 2e\lambda y + f\lambda, \qquad \lambda \neq 0, \tag{$**$}$$

represents the same conic, and this is important to keep in mind to see why they choose such invariants f_1 and f_2.

We know that if we rotate or translate a conic, the following values are invariant (when $\Delta \neq 0$):

$$f_1(\Gamma) = (a+c)\,\Delta^{-\frac{1}{3}}, \tag{1}$$

$$f_2(\Gamma) = (ac - b^2)\,\Delta^{-\frac{2}{3}}, \tag{2}$$

where

$$\Delta = \det \begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix}. \tag{3}$$

Below is a way to do it.

1 For the rotation

I will use the complex rotation matrix, i.e.

$$(x+iy)(\beta+i\gamma) = \beta x - \gamma y + i(\beta y + \gamma x) \;\Rightarrow\; \begin{pmatrix} \beta & -\gamma \\ \gamma & \beta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \beta x - \gamma y \\ \gamma x + \beta y \end{pmatrix}.$$

In order to have a rotation (without scaling) we need to have $\beta^2 + \gamma^2 = 1$. Under the rotation transformation we have

$$x \mapsto \beta x - \gamma y, \qquad y \mapsto \beta y + \gamma x.$$

Plugging the new variables into $(*)$: let (X,Y) be the point after the rotation of (x,y), and suppose it satisfies the general conic equation. We know that $X = \beta x - \gamma y$ and $Y = \beta y + \gamma x$, hence

$$\begin{aligned} p(X,Y) &= aX^2 + 2bXY + cY^2 + 2dX + 2eY + f \\ &= a(\beta x - \gamma y)^2 + 2b(\beta x - \gamma y)(\beta y + \gamma x) + c(\beta y + \gamma x)^2 + 2d(\beta x - \gamma y) + 2e(\beta y + \gamma x) + f. \end{aligned}$$

Simplifying, we get

$$p(X,Y) = Ax^2 + 2Bxy + Cy^2 + 2Dx + 2Ey + F,$$

where

$$\begin{aligned} &A = a\beta^2 + 2b\beta\gamma + c\gamma^2, \quad B = \beta\gamma c + (\beta^2 - \gamma^2)b - \beta\gamma a, \quad C = a\gamma^2 - 2\beta\gamma b + \beta^2 c, \\ &D = \gamma e + \beta d, \qquad\qquad E = \beta e - \gamma d, \qquad\qquad F = f. \end{aligned}$$

Calculation

The quadratic part:

$$\begin{aligned} a(\beta x - \gamma y)^2 &= a\big(\beta^2 x^2 - 2\beta\gamma\, xy + \gamma^2 y^2\big), \\ 2b(\beta x - \gamma y)(\gamma x + \beta y) &= 2b\big(\beta\gamma\, x^2 + (\beta^2 - \gamma^2)\, xy - \beta\gamma\, y^2\big), \\ c(\gamma x + \beta y)^2 &= c\big(\gamma^2 x^2 + 2\beta\gamma\, xy + \beta^2 y^2\big). \end{aligned}$$

Collecting the coefficients of x^2, xy and y^2 gives A, $2B$ and C. The linear part:

$$2d(\beta x - \gamma y) + 2e(\gamma x + \beta y) = 2(\beta d + \gamma e)\, x + 2(\beta e - \gamma d)\, y.$$

I will show that $(a + c)$, $(ac - b^2)$ and Δ are invariant under rotation, i.e.

$$a + c = A + C, \qquad ac - b^2 = AC - B^2$$

and

$$\det \begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix} = \det \begin{pmatrix} A & B & D \\ B & C & E \\ D & E & F \end{pmatrix}.$$

Clearly,

$$A + C = a\beta^2 + 2b\beta\gamma + c\gamma^2 + a\gamma^2 - 2\beta\gamma b + \beta^2 c = a(\beta^2 + \gamma^2) + c(\beta^2 + \gamma^2) = a + c.$$

The other expression, $AC - B^2$: notice that it is the determinant of

$$\begin{pmatrix} A & B \\ B & C \end{pmatrix},$$

so if we prove that

$$\det \begin{pmatrix} a & b \\ b & c \end{pmatrix} = \det \begin{pmatrix} A & B \\ B & C \end{pmatrix},$$

we are done! But that's clear if we look into the transformation matrices (orthogonal matrices of determinant 1):

$$\begin{pmatrix} \beta & \gamma \\ -\gamma & \beta \end{pmatrix} \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} \beta & -\gamma \\ \gamma & \beta \end{pmatrix} = \begin{pmatrix} A & B \\ B & C \end{pmatrix}.$$

It is not an obvious thing to figure out the transformation matrices (at least for me). The right matrix is the same as the matrix we used for the rotation, and the left one is its inverse, but why that's the way I am not sure.

Now for Δ, I will show that we can get from the matrix

$$\begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix} \quad \text{to the matrix} \quad \begin{pmatrix} A & B & D \\ B & C & E \\ D & E & F \end{pmatrix}$$

by multiplying with orthogonal matrices of determinant 1:

$$\begin{pmatrix} \beta & \gamma & 0 \\ -\gamma & \beta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix} \begin{pmatrix} \beta & -\gamma & 0 \\ \gamma & \beta & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} A & B & D \\ B & C & E \\ D & E & F \end{pmatrix}.$$

To see how I came up with these transformation matrices, I included the calculations in the section "The transformation matrices" below.

2 Translation

We need now to show that (1) and (2) are invariant under translations, i.e. under the map

$$(x, y) \mapsto (X, Y) = (x + \beta,\ y + \gamma).$$

We have

$$p(x, y) = ax^2 + 2bxy + cy^2 + 2dx + 2ey + f.$$

Then

$$\begin{aligned} p(X, Y) &= aX^2 + 2bXY + cY^2 + 2dX + 2eY + f \\ &= a(x + \beta)^2 + 2b(x + \beta)(y + \gamma) + c(y + \gamma)^2 + 2d(x + \beta) + 2e(y + \gamma) + f. \end{aligned}$$

Simplifying, we get

$$p(X, Y) = Ax^2 + 2Bxy + Cy^2 + 2Dx + 2Ey + F,$$

where

$$\begin{aligned} &A = a, && B = b, && C = c, \\ &D = a\beta + b\gamma + d, && E = b\beta + c\gamma + e, && F = a\beta^2 + 2b\beta\gamma + c\gamma^2 + 2d\beta + 2e\gamma + f. \end{aligned}$$

> Calculation
>
> Expanding,
>
> $$\begin{aligned} &a(x + \beta)^2 + 2b(x + \beta)(y + \gamma) + c(y + \gamma)^2 + 2d(x + \beta) + 2e(y + \gamma) + f \\ &\quad = ax^2 + 2bxy + cy^2 + 2(a\beta + b\gamma + d)\, x + 2(b\beta + c\gamma + e)\, y \\ &\qquad + a\beta^2 + 2b\beta\gamma + c\gamma^2 + 2d\beta + 2e\gamma + f. \end{aligned}$$

Clearly, we have $A + C = a + c$ and $AC - B^2 = ac - b^2$. Thus the only thing to prove is

$$\det \begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix} = \det \begin{pmatrix} A & B & D \\ B & C & E \\ D & E & F \end{pmatrix}.$$

Again we have transformation matrices (how I figured them out is in the section "The transformation matrices" below):

$$\begin{pmatrix} A & B & D \\ B & C & E \\ D & E & F \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \beta & \gamma & 1 \end{pmatrix} \begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix} \begin{pmatrix} 1 & 0 & \beta \\ 0 & 1 & \gamma \\ 0 & 0 & 1 \end{pmatrix}.$$

Both outer matrices have determinant 1. Thus $\Delta' = \Delta$.

3 Other things to consider

We have that $a+c$, $ac-b^2$ and Δ are all invariant under rotation and translation. The question is: why these specific formulas for f_1, f_2? It turns out we need to consider other things, like the fact that $(*)$ and $(**)$ represent the same conic, i.e. if we scale all coefficients (not (x,y)) we should have the same values for f_1, f_2:

$$\lambda p(x,y) = \lambda\left(ax^2+2bxy+cy^2+2dx+2ey+f\right),$$

and that will not change the conic. Hence we should have the same f_1 and f_2 values after multiplying the equation by a nonzero number! In this case we have

$$A=\lambda a,\quad B=\lambda b,\quad C=\lambda c,\quad D=\lambda d,\quad E=\lambda e,\quad F=\lambda f.$$

Then

$$\Delta' = \det\begin{pmatrix} A & B & D\\ B & C & E\\ D & E & F\end{pmatrix} = \det\begin{pmatrix} \lambda a & \lambda b & \lambda d\\ \lambda b & \lambda c & \lambda e\\ \lambda d & \lambda e & \lambda f\end{pmatrix} = \lambda^3 \det\begin{pmatrix} a & b & d\\ b & c & e\\ d & e & f\end{pmatrix} = \lambda^3\Delta.$$

Then

$$\frac{\lambda a+\lambda c}{\sqrt[3]{\lambda^3\Delta}} = (a+c)\,\Delta^{-\frac{1}{3}}, \qquad \frac{\lambda^2 ac-\lambda^2 b^2}{\sqrt[3]{\lambda^6\Delta^2}} = (ac-b^2)\,\Delta^{-\frac{2}{3}}.$$

Further, we need the set f_1, f_2 to be complete, i.e. for different conics we need f_1, f_2 to output different values. I guess that's why we need to include Δ in f_1, f_2, since it depends on all the coefficients. I mean, why not $f=\frac{(a+c)^2}{ac-b^2}$? It is invariant too, but it does not involve d, e, f, so it cannot distinguish circles of different radii: every circle $x^2+y^2=r^2$ gives $f=4$.

4 Scaling

We will show that f_1, f_2 are not invariant under scaling. To do that, suppose (X,Y) is the point after scaling (x,y), satisfying

$$p(X,Y) = aX^2+2bXY+cY^2+2dX+2eY+f.$$

Assume the scaling factor to be λ, i.e. $X=\lambda x$ and $Y=\lambda y$. Then

$$p(X,Y) = a\lambda^2x^2+2b\lambda^2xy+c\lambda^2y^2+2d\lambda x+2e\lambda y+f.$$

Then the new matrix is

$$\begin{pmatrix} a\lambda^2 & b\lambda^2 & \lambda d\\ b\lambda^2 & c\lambda^2 & \lambda e\\ \lambda d & \lambda e & f\end{pmatrix}.$$

It is easy to see that we can reach this matrix from

$$\begin{pmatrix} a & b & d\\ b & c & e\\ d & e & f\end{pmatrix}$$

by multiplying the first row by λ, then the first column, then the second row, and lastly the second column. Thus the determinant would be

$$\Delta' = \lambda^4 \Delta.$$

Hence, since $A + C = \lambda^2(a + c)$,

$$f_1(\Gamma') = \frac{A + C}{\sqrt[3]{\Delta'}} = \frac{\lambda^2(a + c)}{\sqrt[3]{\lambda^4 \Delta}} = \lambda^{\frac{2}{3}} f_1(\Gamma) \neq f_1(\Gamma)$$

(unless $\lambda = \pm 1$), and in the same way $f_2(\Gamma') = \lambda^{\frac{4}{3}} f_2(\Gamma)$.

5 The transformation matrices

We look into the simple row/column operations: adding a multiple of a row/column to another will not affect the determinant, while multiplying a row or a column by a scalar will scale the determinant. We can do the following, using simple row/column operations with multiplying a row/column by a constant.

The rotation.

Calculation

$$\begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix} \xrightarrow{\beta R_2 - \gamma R_1 \mapsto R_2} \begin{pmatrix} a & b & d \\ \beta b - \gamma a & \beta c - \gamma b & \beta e - \gamma d \\ d & e & f \end{pmatrix}.$$

That would multiply the determinant by β.

$$\xrightarrow{\beta C_2 - \gamma C_1 \mapsto C_2} \begin{pmatrix} a & \beta b - \gamma a & d \\ \beta b - \gamma a & \beta^2 c - 2\beta\gamma b + \gamma^2 a & \beta e - \gamma d \\ d & \beta e - \gamma d & f \end{pmatrix}.$$

Again this will multiply the determinant by β, so the matrix we have will have β^2 times the determinant of the original.

$$\xrightarrow{\frac{1}{\beta} R_1 + \frac{\gamma}{\beta} R_2 \mapsto R_1} \begin{pmatrix} \beta a + \gamma b & (\beta^2 - \gamma^2) b + \beta\gamma(c - a) & \beta d + \gamma e \\ \beta b - \gamma a & \beta^2 c - 2\beta\gamma b + \gamma^2 a & \beta e - \gamma d \\ d & \beta e - \gamma d & f \end{pmatrix},$$

using $\frac{a}{\beta} - \frac{\gamma^2}{\beta} a = \frac{\beta^2}{\beta} a = \beta a$ and $1 - 2\gamma^2 = \beta^2 - \gamma^2$. This will multiply the determinant by β^{-1}, so the result is β times the original determinant.

$$\xrightarrow{\frac{1}{\beta} C_1 + \frac{\gamma}{\beta} C_2 \mapsto C_1} \begin{pmatrix} \beta^2 a + 2\beta\gamma b + \gamma^2 c & (\beta^2 - \gamma^2) b + \beta\gamma(c - a) & \beta d + \gamma e \\ (\beta^2 - \gamma^2) b + \beta\gamma(c - a) & \beta^2 c - 2\beta\gamma b + \gamma^2 a & \beta e - \gamma d \\ \beta d + \gamma e & \beta e - \gamma d & f \end{pmatrix},$$

and this multiplies the determinant by β^{-1} again, so the determinant is back to the original one. For the $(2, 1)$ entry:

$$\frac{\beta b - \gamma a}{\beta} + \frac{\gamma}{\beta}(\beta^2 c - 2\beta\gamma b + \gamma^2 a) = (1 - 2\gamma^2) b + \beta\gamma c - \frac{\gamma(1 - \gamma^2)}{\beta} a = (\beta^2 - \gamma^2) b + \beta\gamma(c - a).$$

This is the matrix with entries A, B, C, D, E, F.
Row operations mean we multiply from the left by the identity matrix that suffered the same row operation, i.e. to capture $\beta R_2 - \gamma R_1 \mapsto R_2$ we have

$$\begin{pmatrix} 1 & 0 & 0 \\ -\gamma & \beta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix} = \begin{pmatrix} a & b & d \\ \beta b - \gamma a & \beta c - \gamma b & \beta e - \gamma d \\ d & e & f \end{pmatrix}.$$

Thus in total from the left (row operations) we have

$$\begin{pmatrix} \beta^{-1} & \gamma\beta^{-1} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ -\gamma & \beta & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \beta^{-1} - \gamma^2\beta^{-1} & \gamma & 0 \\ -\gamma & \beta & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \beta & \gamma & 0 \\ -\gamma & \beta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

For column operations we multiply from the right by the identity matrix that suffered the same column operation, i.e. to capture $\beta C_2 - \gamma C_1 \mapsto C_2$ we multiply

$$\begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix} \begin{pmatrix} 1 & -\gamma & 0 \\ 0 & \beta & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} a & \beta b - \gamma a & d \\ b & \beta c - \gamma b & e \\ d & \beta e - \gamma d & f \end{pmatrix}.$$

The translation. We need

$$\det \begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix} = \det \begin{pmatrix} a & b & a\beta + b\gamma + d \\ b & c & b\beta + c\gamma + e \\ a\beta + b\gamma + d & b\beta + c\gamma + e & a\beta^2 + 2b\beta\gamma + c\gamma^2 + 2d\beta + 2e\gamma + f \end{pmatrix}.$$

We add multiples of rows/columns to other rows/columns, and that will not affect the determinant.

Calculation

$$\begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix} \xrightarrow{\beta C_1 + C_3 \mapsto C_3} \begin{pmatrix} a & b & d + \beta a \\ b & c & e + \beta b \\ d & e & f + \beta d \end{pmatrix} \xrightarrow{\gamma C_2 + C_3 \mapsto C_3} \begin{pmatrix} a & b & d + \beta a + \gamma b \\ b & c & e + \beta b + \gamma c \\ d & e & f + \beta d + \gamma e \end{pmatrix}.$$

And

$$\xrightarrow{\beta R_1 + R_3 \mapsto R_3} \begin{pmatrix} a & b & d + \beta a + \gamma b \\ b & c & e + \beta b + \gamma c \\ d + \beta a & e + \beta b & f + 2\beta d + \gamma e + \beta^2 a + \beta\gamma b \end{pmatrix}.$$

Lastly,

$$\xrightarrow{\gamma R_2 + R_3 \mapsto R_3} \begin{pmatrix} a & b & d + \beta a + \gamma b \\ b & c & e + \beta b + \gamma c \\ d + \beta a + \gamma b & e + \beta b + \gamma c & f + 2\beta d + 2\gamma e + \beta^2 a + 2\beta\gamma b + \gamma^2 c \end{pmatrix}.$$

Doing the same as for the rotation gets us the transformation matrices.

6 Further investigations

I found out where the matrix with determinant Δ comes from. Prof. Candelori suggested to complete the square, and I found something interesting when I did that:

$$ax^2 + 2bxy + cy^2 + 2dx + 2ey + f = a\left(x + \frac{b}{a}y + \frac{d}{a}\right)^2 + \left(\frac{ac - b^2}{a}\right)\left(y + \frac{ae - bd}{ac - b^2}\right)^2 + \frac{\Delta}{ac - b^2}$$

(for $a \neq 0$ and $ac - b^2 \neq 0$). So at the end we get Δ.

> Calculation
>
> $$\begin{aligned}
> &ax^2 + 2bxy + cy^2 + 2dx + 2ey + f \\
> &= a\left(x^2 + \frac{2b}{a}xy\right) + cy^2 + 2dx + 2ey + f \\
> &= a\left(x + \frac{b}{a}y\right)^2 - \frac{b^2}{a}y^2 + cy^2 + 2d\left(x + \frac{b}{a}y\right) - \frac{2db}{a}y + 2ey + f \\
> &= a\left[\left(x + \frac{b}{a}y\right)^2 + \frac{2d}{a}\left(x + \frac{b}{a}y\right)\right] + \left(\frac{ac - b^2}{a}\right)y^2 + 2\,\frac{ae - bd}{a}\,y + f \\
> &= a\left[x + \frac{b}{a}y + \frac{d}{a}\right]^2 - \frac{d^2}{a} + \left(\frac{ac - b^2}{a}\right)\left(y^2 + 2\,\frac{ae - bd}{ac - b^2}\,y\right) + f \\
> &= a\left(x + \frac{b}{a}y + \frac{d}{a}\right)^2 + \left(\frac{ac - b^2}{a}\right)\left(y + \frac{ae - bd}{ac - b^2}\right)^2 - \frac{d^2}{a} - \frac{(ae - bd)^2}{a(ac - b^2)} + f \\
> &= a\left(x + \frac{b}{a}y + \frac{d}{a}\right)^2 + \left(\frac{ac - b^2}{a}\right)\left(y + \frac{ae - bd}{ac - b^2}\right)^2 + \frac{\Delta}{ac - b^2}.
> \end{aligned}$$
>
> For the last step: over the common denominator $a(ac - b^2)$, the numerator of the constant is $af(ac - b^2) - d^2(ac - b^2) - (ae - bd)^2 = a\big(acf - ae^2 - b^2f + 2bde - cd^2\big) = a\Delta$.

7 The conic definitions

Ellipse. We have two points, the foci $F_1(h_1, k_1)$ and $F_2(h_2, k_2)$. The ellipse is all points $P(x, y)$ in $\mathbb{R}^2$ such that the sum of distances $d_1 + d_2 = D$ is a fixed number D, where $d_1 = PF_1 = \sqrt{(x - h_1)^2 + (y - k_1)^2}$ and $d_2 = PF_2 = \sqrt{(x - h_2)^2 + (y - k_2)^2}$. To get the equation we need to square both sides twice, which gives

$$0 = D^4 - 2D^2(d_1^2 + d_2^2) + \big((2x - (h_1 + h_2))(h_2 - h_1) + (2y - (k_1 + k_2))(k_2 - k_1)\big)^2.$$

> Calculation

$$
\begin{aligned}
d_1 + d_2 &= D \\
(d_1 + d_2)^2 &= D^2 \\
d_1^2 + 2d_1d_2 + d_2^2 &= D^2 \\
2d_1d_2 &= D^2 - (d_1^2 + d_2^2) \\
4d_1^2d_2^2 &= D^4 - 2D^2(d_1^2 + d_2^2) + (d_1^4 + 2d_1^2d_2^2 + d_2^4) \\
0 &= D^4 - 2D^2(d_1^2 + d_2^2) + (d_1^2 - d_2^2)^2 \\
0 &= D^4 - 2D^2(d_1^2 + d_2^2) + \big((x-h_1)^2 - (x-h_2)^2 + (y-k_1)^2 - (y-k_2)^2\big)^2 \\
0 &= D^4 - 2D^2(d_1^2 + d_2^2) + \big((2x - (h_1+h_2))(h_2-h_1) + (2y - (k_1+k_2))(k_2-k_1)\big)^2,
\end{aligned}
$$

using $(x-h_1)^2 - (x-h_2)^2 = (h_2-h_1)(2x-h_1-h_2)$, and the same for y.

Evaluate $ac - b^2$, where a, c, b are the coefficients of x^2, y^2 and (half of) xy respectively:

$$a = -4D^2 + 4(h_2-h_1)^2, \qquad c = -4D^2 + 4(k_2-k_1)^2, \qquad b = 4(h_2-h_1)(k_2-k_1),$$

then

$$ac - b^2 = 16\Big[D^4 - D^2\big((h_2-h_1)^2 + (k_2-k_1)^2\big) + (h_2-h_1)^2(k_2-k_1)^2 - (h_2-h_1)^2(k_2-k_1)^2\Big].$$

Notice that $\sqrt{(h_2-h_1)^2 + (k_2-k_1)^2}$ is the distance between the foci, usually called $2C$, because the midpoint between the two foci is called the center and C is the distance from one focus to the center. Thus, using C (half the distance between the two foci), we have

$$ac - b^2 = 16D^4 - 64D^2C^2 = 16D^2(D^2 - 4C^2) > 0,$$

since $D^2 - 4C^2 = 4B^2$, where B is the half minor axis length. An ellipse has two axes: the half length of the major axis is usually called A, and B is the half minor axis length. In the standard form they appear as $\frac{x^2}{B^2} + \frac{y^2}{A^2} = 1$ with $A > B$, and we have $A^2 = B^2 + C^2$ and $D = 2A$ (if we assumed the major axis is vertical; for horizontal, A^2 is the denominator of x^2).

Hyperbola. Again we have two foci $F_1(h_1, k_1)$ and $F_2(h_2, k_2)$, and the hyperbola is all the (x, y) such that $|d_1 - d_2| = D$. Simplifying, we get the same formula for $ac - b^2$:

$$ac - b^2 = 16D^4 - 64D^2C^2 = 16D^2(D^2 - 4C^2) < 0,$$

since here we have $C^2 = A^2 + B^2$, where C is the same as for the ellipse, in the standard form $\frac{x^2}{A^2} - \frac{y^2}{B^2} = 1$, and $D = 2A$; thus $D^2 - 4C^2 = -4B^2 < 0$.

Parabola. Finally, the parabola is the set of points (x, y) that are equidistant from a given point (h, k) and a line $rx + ty + s = 0$, i.e.

$$\sqrt{(x-h)^2 + (y-k)^2} = \frac{|rx + ty + s|}{\sqrt{r^2 + t^2}}.$$

Simplifying, we get $a = t^2$, $c = r^2$, $b = -rt$ in this case, and then $ac - b^2 = 0$.

Calculation

Squaring and multiplying by $r^2 + t^2$:

$$(r^2 + t^2)\big((x-h)^2 + (y-k)^2\big) - (rx + ty + s)^2 = 0,$$

whose quadratic terms are $(r^2 + t^2)(x^2 + y^2) - (r^2x^2 + 2rt\,xy + t^2y^2) = t^2x^2 - 2rt\,xy + r^2y^2$.

8 If the general polynomial factors (reducible), then $\Delta = 0$

If $ax^2 + 2bxy + cy^2 + 2dx + 2ey + f$ factors,

$$ax^2 + 2bxy + cy^2 + 2dx + 2ey + f = (\alpha_1 x + \beta_1 y + \gamma_1)(\alpha_2 x + \beta_2 y + \gamma_2),$$

then simplifying we get

$$a = \alpha_1\alpha_2, \quad 2b = \alpha_1\beta_2 + \beta_1\alpha_2, \quad c = \beta_1\beta_2, \quad 2d = \alpha_1\gamma_2 + \alpha_2\gamma_1, \quad 2e = \beta_1\gamma_2 + \beta_2\gamma_1, \quad f = \gamma_1\gamma_2.$$

Thus we need to show that

$$\det \begin{pmatrix} 2\alpha_1\alpha_2 & \alpha_1\beta_2 + \alpha_2\beta_1 & \alpha_1\gamma_2 + \alpha_2\gamma_1 \\ \alpha_1\beta_2 + \alpha_2\beta_1 & 2\beta_1\beta_2 & \beta_1\gamma_2 + \beta_2\gamma_1 \\ \alpha_1\gamma_2 + \alpha_2\gamma_1 & \beta_1\gamma_2 + \beta_2\gamma_1 & 2\gamma_1\gamma_2 \end{pmatrix} = 0,$$

and that's the case.

Calculation

With $u = (\alpha_1, \beta_1, \gamma_1)^T$ and $w = (\alpha_2, \beta_2, \gamma_2)^T$, this matrix is $uw^T + wu^T$. Every column of it is a combination of u and w, so its rank is at most 2 and its determinant is 0.

9 Eliminating the degree 1 and constant terms

While Jack was presenting, Prof. Candelori posed the question: can we eliminate the degree 1 and constant terms of

$$p(x, y) = ax^2 + 2bxy + cy^2 + 2dx + 2ey + f$$

using rotations and translations? I was thinking it is yes, but I think I am wrong, since if we have $ax^2 + 2bxy + cy^2$, all terms of degree 2, then by completing the square we end up with

$$a\left(x + \frac{b}{a}y\right)^2 + \left(c - \frac{b^2}{a}\right)y^2 = 0,$$

which is either a single point or two lines (real or complex). Also, its matrix has last row $(0, 0, 0)$, so its Δ is 0, while rotations and translations do not change Δ; so when $\Delta \neq 0$ the answer is no. Note that the map $x \mapsto x + \frac{b}{a}y$, $y \mapsto y$ is not a rotation or translation; it is a shear.